

FUNDAMENTALS OF DATA MINING

Lecture 5

Clustering

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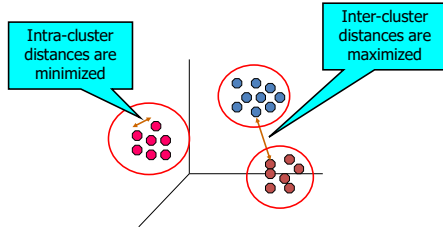
This Lecture...

- Cluster Analysis: Basic Concepts and Algorithms
- Types of Clusterings
- K-means Clustering

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What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups

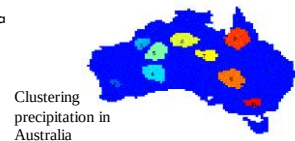


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Applications of Cluster Analysis

- **Understanding**
 - Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations
- **Summarization**
 - Reduce the size of large data sets

	Discovered Clusters	Industry Group
1	Applied-Mat-DOWN, Bio-Network-Tools-UP, GM-DOWN, Calabrese-Sys-DOWN, CSCO-DOWN, JP-DOWN, EMC-Corps-DOWN, INTEL-DOWN, LSI-logic-DOWN, Microw-Tech-DOWN, Texas-Instr-DOWN, Telford-Ind-DOWN, Nall-Semiconductor-DOWN, Novus-DOWN, Siga-DOWN, Sun-DOWN	Technology 1-DOWN
2	Amtek-Corps-DOWN, Autodesk-DOWN, DEC-DOWN, ADY-Micro-Device-DOWN, Andrew-Corp-DOWN, Computer-Agency-DOWN, Corvus-Corp-DOWN, Compaq-DOWN, EMC-Corp-DOWN, Gen-Int-DOWN, Minolta-DOWN, Microsoft-DOWN, Scientific-Air-DOWN	Technology 2-DOWN
3	Farmat-Mat-DOWN, Fed-Home-Land-DOWN, MENA-Corp-DOWN, Morgan-Stanley-DOWN	Financial-DOWN
4	Baker-Hughes-UP, Dresser-Ind-UP, Halliburton-Ind-UP, Louisiana-Land-UP, Phillips-Petroleum-UP, Schlumberger-UP	Oil-UP



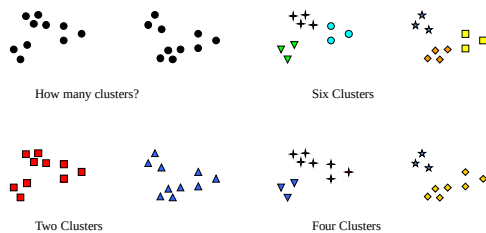
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What is NOT Cluster Analysis?

- **Supervised Learning**
 - Have class label information
- **Simple segmentation**
 - Dividing students into different registration groups alphabetically, by last name
- **Results of a query**
 - Groupings are a result of an external specification

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Notion of a Cluster can be Ambiguous



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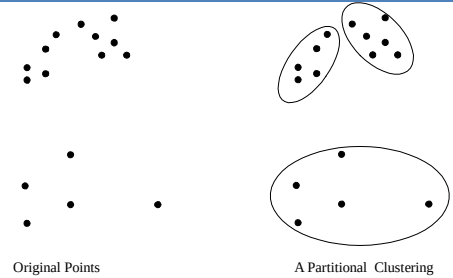
Types of Clusterings

Hierarchical vs. Partitional Clusterings

- Important distinction between hierarchical and partitional sets of clusters,
 - ▣ Partitional Clustering
 - A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
 - ▣ Hierarchical clustering
 - A set of nested clusters organized as a hierarchical tree

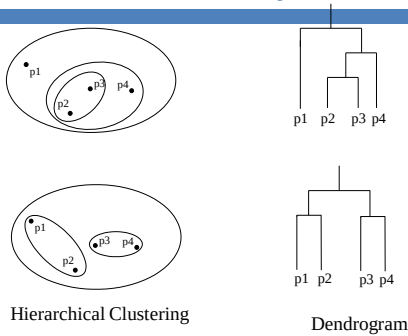
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Partitional Clustering



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Hierarchical Clustering



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Types of Clusterings

- Exclusive versus non-exclusive
 - ▣ In non-exclusive clusterings, points may belong to multiple clusters.
 - ▣ Can represent multiple classes or 'border' points
- Fuzzy versus non-fuzzy
 - ▣ In fuzzy clustering, a point belongs to every cluster with some weight between 0 and 1
 - ▣ Weights must sum to 1
 - ▣ Probabilistic clustering has similar characteristics
- Partial versus complete
 - ▣ In some cases, we only want to cluster some of the data

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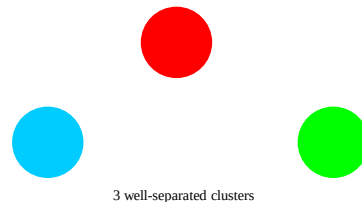
Types of Clusters

- Well-separated clusters
- Center-based clusters
- Contiguous clusters
- Density-based clusters
- Property or Conceptual clusters

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Types of Clusters: Well-Separated

- Well-Separated Clusters:
 - ▣ Each point in a cluster is closer (or more similar) to all of the points in its cluster than to any point in another cluster.



3 well-separated clusters

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Types of Clusters: Center-Based

- Center-based (Prototype-based)
 - ▣ Each point is closer to the center of its cluster than to the center of any other cluster.
 - ▣ The center of a cluster is often a **centroid**, the average of all the points in the cluster, or a **medoid**, the most "representative" point of a cluster



4 center-based clusters

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Types of Clusters: Contiguity-Based

- Contiguous Cluster (Nearest neighbor or Transitive)
 - ▣ Each point is closer (or more similar) to at least one point in its cluster than to any point in another cluster.

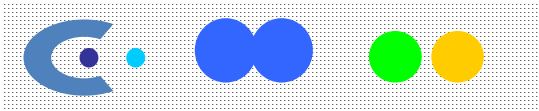


8 contiguous clusters

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Types of Clusters: Density-Based

- Density-based
 - ▣ A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
 - ▣ Used when the clusters are irregular or intertwined, and when noise and outliers are present.



6 density-based clusters

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Types of Clusters: Conceptual Clusters

- Shared Property or Conceptual Clusters
 - ▣ Finds clusters that share some common property or represent a particular concept. (Points in the intersection of the circles belong to both)



2 Overlapping Circles

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Clustering Algorithms

- K-means and its variants
- Hierarchical clustering
- Density-based clustering

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K-means Clustering

- Partitional clustering approach
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- Number of clusters, K , must be specified
- The basic algorithm is very simple

- 1: Select K points as the initial centroids.
- 2: **repeat**
- 3: Form K clusters by assigning all points to the closest centroid.
- 4: Recompute the centroid of each cluster.
- 5: **until** The centroids don't change

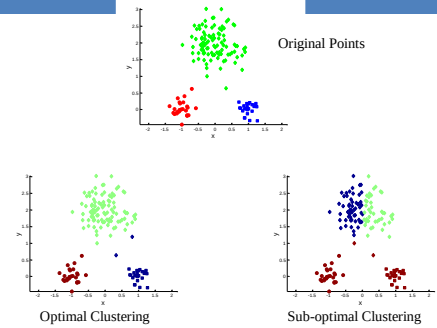
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K-means Clustering – Details

- Initial centroids are often chosen randomly.
 - ▣ Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- 'Closeness' is measured by Euclidean distance, Manhattan Distance, cosine similarity, correlation, etc.
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
 - ▣ Often the stopping condition is changed to 'Until relatively few points change clusters'

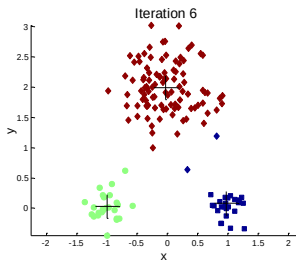
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Two different K-means Clusterings



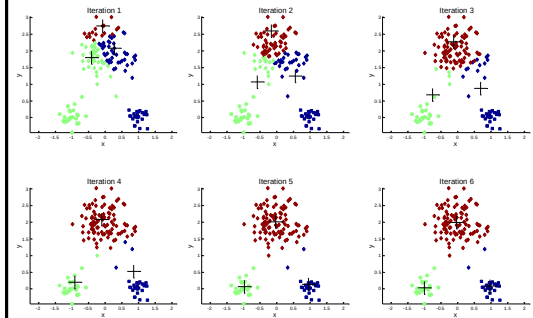
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Importance of Choosing Initial Centroids



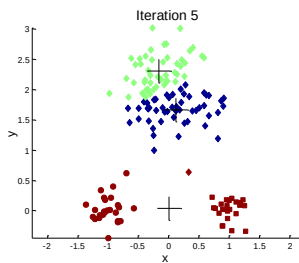
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Importance of Choosing Initial Centroids



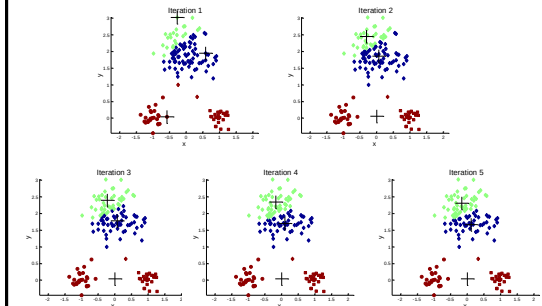
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Importance of Choosing Initial Centroids ...



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Importance of Choosing Initial Centroids ...



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Evaluating K-means Clusters

- Most common measure is Sum of Squared Error (SSE)
 - ▣ For each point, the error is the distance to the nearest cluster
 - ▣ To get SSE, we square these errors and sum them.

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} \text{dist}^2(m_i, x)$$

- ▣ x is a data point in cluster C_i and m_i is the representative point for cluster C_i
 - can show that m_i corresponds to the center (mean) of the cluster
- ▣ Given two clusters, we can choose the one with the smallest error
- ▣ One easy way to reduce SSE is to increase K , the number of clusters
 - A good clustering with smaller K can have a lower SSE than a poor clustering with higher K

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Solutions to Initial Centroids Problem

- Multiple runs
 - ▣ Helps, but probability is not on your side
- Select more than k initial centroids and then select among these initial centroids
 - ▣ Select most widely separated
- Sample and use hierarchical clustering to determine initial centroids
- Postprocessing

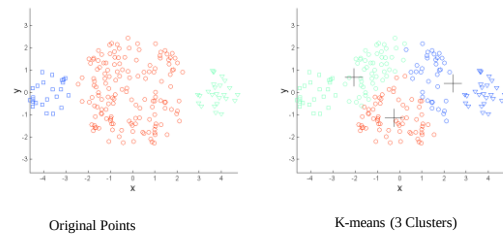
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Limitations of K-means

- K-means has problems when clusters are of differing
 - ▣ Sizes
 - ▣ Densities
 - ▣ Non-globular shapes
- K-means has problems when the data contains outliers.

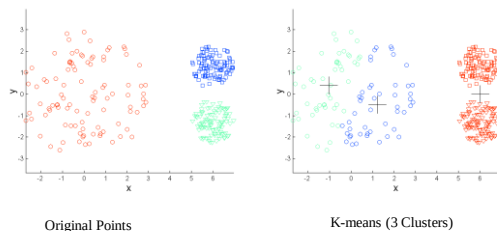
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Limitations of K-means: Differing Sizes



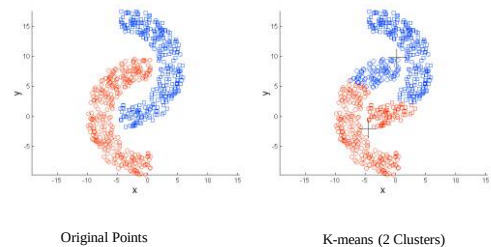
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Limitations of K-means: Differing Density



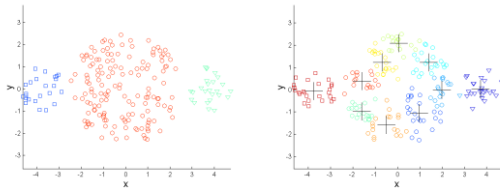
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Limitations of K-means: Non-globular Shapes



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Overcoming K-means Limitations



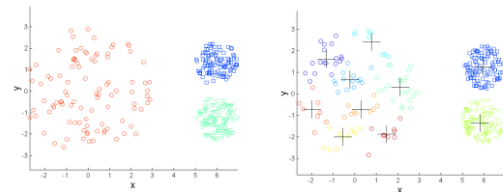
Original Points

K-means Clusters

One solution is to use many clusters.
Find parts of clusters, but need to put together.

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Overcoming K-means Limitations

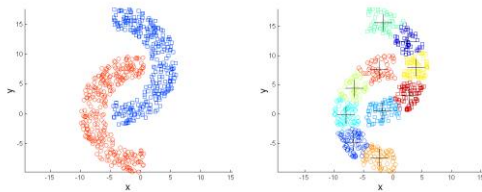


Original Points

K-means Clusters

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Overcoming K-means Limitations



Original Points

K-means Clusters

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Example

- Cluster the following points (with (x,y) representing locations) into three clusters.
Initial cluster centers are A1, A4, A7

Data Points	x	y
A1	2	10
A2	2	5
A3	8	4
A4	5	8
A5	7	5
A6	6	4
A7	1	2
A8	4	9

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Distance Measures

- Euclidean distance (Statistical Distance)

$$d(p,q) = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2 + \dots + (q_n - p_n)^2}$$

$$= \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

- Manhattan distance

$$d(x,y) = \sum_{i=1}^n |x_i - y_i|$$

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